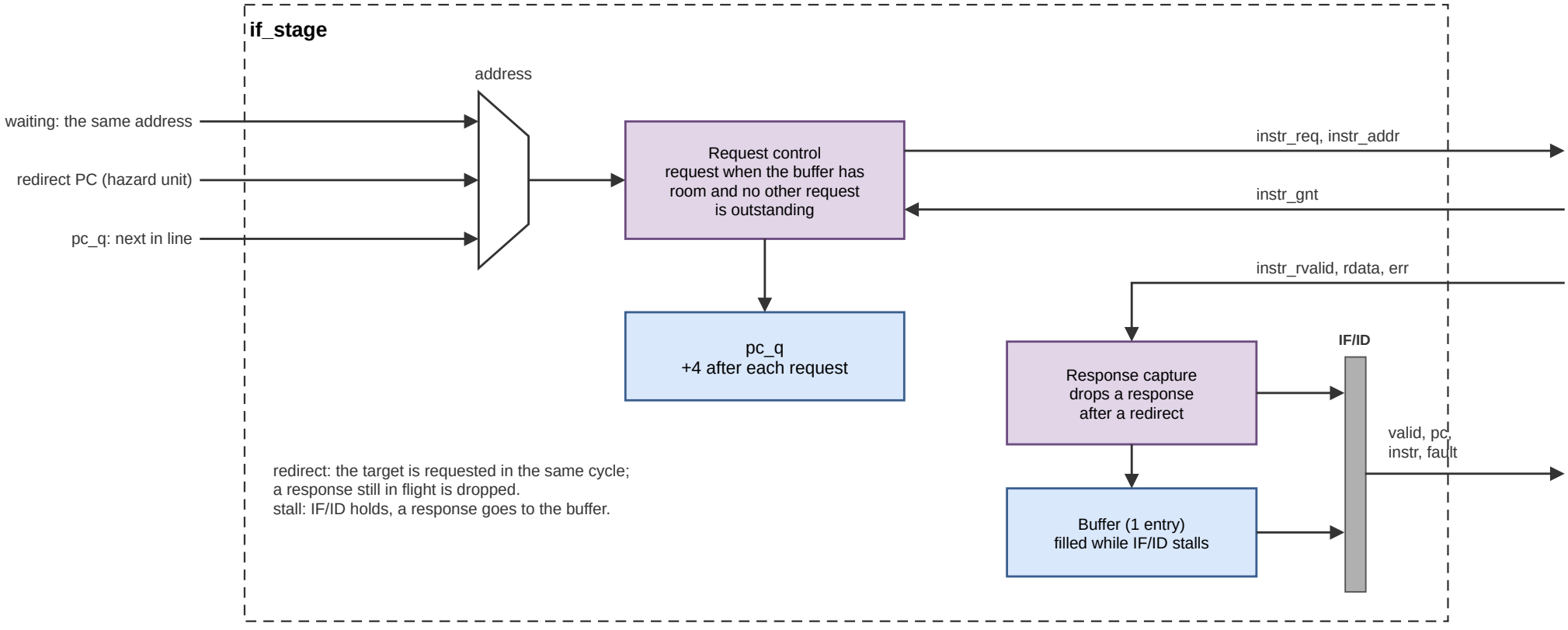
if_stage
address
waiting: the same address
redirect PC (hazard unit)
pc_q: next in line
Request control
request when the buffer has
room and no other request
is outstanding
instr_req, instr_addr
instr_gnt
instr_rvalid, rdata, err
pc_q
+4 after each request
IF/ID
Response capture
drops a response
after a redirect
valid, pc,
instr, fault
Buffer (1 entry)
filled while IF/ID stalls
redirect: the target is requested in the same cycle;
a response still in flight is dropped.
stall: IF/ID holds, a response goes to the buffer.